

# Data Driven Systems

Bibliographic details

## Introduction

The purpose is to study in greater detail the core concepts and principles presented in this course, going beyond the perspective provided by the practical assignments. These assignments offered hands-on experience by focusing on selected topics within the field; however, they provided only a partial view of the broader domain, as they were centered on specific technologies, methods and design choices. The aim is to bridge existing gaps by studying subjects that were not covered in practical work and by situating the subjects covered in the practical work within a more general and conceptual framework of the field. It is expected to develop a comprehensive understanding of the subject area, enabling students to make informed decisions when designing and implementing real-world systems. This includes the ability to critically assess AI-generated solutions and leverage them more effectively.

## Spatial Data Management

The management and processing of spatial data have been active areas of research and development for several decades and there have been well-established solutions since the early 2000s. This is supported by 1) clearly defined standards by the Open Geospatial Consortium (OGC); and 2) widely used libraries such as GEOS, GDAL or PROJ, implementing the fundamental methods and operations for spatial data processing and serving as the building blocks of many contemporary tools like PostGIS, Shapely and GeoPandas.

Recent developments in the field focus less on introducing new fundamental methods and more on the design of interfaces and integration layers that align spatial data processing with modern database systems, programming languages, and data analytics platforms. As a result, a solid understanding of the underlying concepts and principles enables students to work effectively with most of the spatial data tools in use today, as well as with those likely to emerge in the future. The bibliography supporting the study of this topic is:

*Nikos Mamoulis. 2012. Spatial data management. Synthesis Lectures on Data Management, Morgan & Claypool publishers.*

- *Chapter 1. Introduction*
- *Chapter 2. Spatial data*
- *Chapter 3. Indexing, up to and including section 3.3.2.*

Current research is focused on specific topics like real-time processing, motivated by applications such as autonomous driving and other data-intensive, spatially aware systems, as illustrated in several paper presentations discussed during the semester.

Note that the designation spatial data encompasses all data tied to a position in space, whereas geographical data is targeted to Earth-based locations. For example, spatial data tools can be used to represent and analyze the spatial evolution of cells in microscopic image sequences.

## Multidimensional Data Modeling, Data Warehousing and OLAP

Multidimensional data modeling and data warehousing emerged in the 1990s as highly structured approaches for decision support systems and analytical processing, addressing limitations of operational databases for handling

complex analytical queries (OLAP). Early data warehouses relied on centralized architectures and well-defined schemas, such as star and snowflake schemas, to organize data into facts, dimensions, and hierarchies optimized for OLAP operations. Although modern data architectures can provide greater flexibility, traditional data warehousing approaches remain relevant and continue to be the most appropriate solution for many use cases. The bibliography supporting the study of this topic is:

*Christian Jensen, Torben Pedersen and Christian Thomsen. 2010. Multidimensional Databases and Data Warehousing. Synthesis Lectures on Data Management, Morgan & Claypool publishers.*

- *Chapter 1. Introduction*
- *Chapter 2. Fundamental Concepts*

## **Data Lakes and Data Lakehouses**

Over time, increasing data volume, variety, and velocity led to the adoption of distributed and more flexible architectures, as well as alternative integration strategies and storage paradigms. While traditional data warehouses emphasized strong structure, data consistency, and performance for predefined analytical workloads, subsequent approaches such as data lakes and, more recently, data lakehouses, introduced greater flexibility by accommodating heterogeneous data and evolving schemas. Despite these changes, multidimensional modeling remains a foundational concept to the design of modern analytical systems. The reference for this topic is the following paper:

*Michael Armbrust, Tathagata Das, Liwen Sun, Burak Yavuz, Shixiong Zhu, Mukul Murthy, Joseph Torres, Herman van Hovell, Adrian Ionescu, Alicja Łuszczak, Michał Świtakowski, Michał Szafranski, Xiao Li, Takuya Ueshin, Mostafa Mokhtar, Peter Boncz, Ali Ghodsi, Sameer Paranjpye, Pieter Senster, Reynold Xin, and Matei Zaharia. 2020. Delta lake: high-performance ACID table storage over cloud object stores. Proc. VLDB Endow. 13, 12 (August 2020), 3411–3424.*

<https://doi.org/10.14778/3415478.3415560>

This paper is a foundational contribution to the emergence of data lakehouse architectures and to understanding recent developments in large-scale analytical data management. It introduces key design principles like ACID transaction support, schema enforcement and evolution, time travel and scalable metadata management, to address several limitations of data lakes built on cloud object storage. The concepts and mechanisms presented in this work are not specific to Delta Lake alone but are largely shared by alternative open table formats, such as Apache Iceberg and Apache Hudi, which provide comparable guarantees and functionality through different implementations. As such, Delta Lake and Delta tables represent well the recent developments on cloud-native analytics platforms.

## **Endnotes**

- The exam does not cover all topics discussed throughout the semester or listed in the course slides.
- Lecture slides are not intended to serve as study material.
- Consider the exercises at the end of each chapter in *“Multidimensional Databases and Data Warehousing”* as part of their study process.